



Jérôme Eertmans

Ph.D. student in Differentiable Ray Tracing for Telecommunications

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📖 Google Scholar

Education

Université catholique de Louvain, Ph.D. in Electrical engineering Sept 2021 – July 2026
Thesis title: Differentiable Ray Tracing for Radio Propagation Modeling

Université catholique de Louvain, M.Sc. in Electro-mechanical engineering Sept 2019 – June 2021
Majored in mechatronics.

- I undertook a variety of course topics: telecommunications, open source software, numerical simulations, digital design of ICs, machine learning, mining pattern in data, etc., for a total of 112% workload.
- Graduated Magna Cum Laude.

Experience

Teaching Assistant, Université catholique de Louvain – Louvain-la-Neuve, Belgium Sept 2021 – present
• Teaching duties (practical sessions, evaluation, ...): see list [here](#)
• Research on my Ph.D. subject, under the supervision of Profs. Claude Oestges and Laurent Jacques.

Visiting Researcher, University of Bologna – Bologna, Italy Sept 2024 – Dec 2024
Visiting Prof. Vittorio Degli-Esposti's lab and pursuing research on Machine Learning applied to Ray Tracing for Radio Propagation.

For additional experiences, see my [LinkedIn profile](#).

Languages

French: Native

English: B2+/C1

Dutch: B1

Italian: A2

Japanese: Basics

Projects

DiffeRT

Python package with backend written in Rust.

- Built on top of [JAX](#) for high-performance and differentiable code
- Thoroughly tested and well documented
- Fully open access, along with its 2D variant [DiffeRT2D](#)

Manim Slides

Presentation tool I developed to produce higher-quality presentations at conferences.

- Most of my presentations are created using this tool and can be viewed on [my personal website](#)
- Used by other researchers to disseminate their research, with examples available in the [documentation](#)

More projects

You may find an extended list of my projects and contributions on my [GitHub profile](#) or on the [projects page](#) of my website.

Publications

- Transform-Invariant Generative Ray Path Sampling for Efficient Radio Propagation Modeling** Mar 2026
Jérôme Eertmans, Enrico Maria Vitucci, Vittorio Degli-Esposti, Nicola Di Cicco, Laurent Jacques, Claude Oestges
arxiv.org/abs/2603.01655 (arXiv preprint, submitted to npj Wireless Technology)
- Fast, Differentiable, GPU-Accelerated Ray Tracing for Multiple Diffraction and Reflection Paths** Oct 2025
Jérôme Eertmans, Sophie Lequeu, Benoît Legat, Laurent Jacques, Claude Oestges
arxiv.org/abs/2410.14535 (arXiv preprint, accepted at EuCAP 2026)
- Towards Generative Ray Path Sampling for Faster Point-to-Point Ray Tracing** Oct 2024
Jérôme Eertmans, Nicola Di Cicco, Claude Oestges, Laurent Jacques, Enrico Maria Vitucci, Vittorio Degli-Esposti
[10.1109/ICMLCN64995.2025.11140249](https://doi.org/10.1109/ICMLCN64995.2025.11140249) (2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN))
- Comparing Differentiable and Dynamic Ray Tracing - Introducing the Multipath Lifetime Map** Oct 2024
Jérôme Eertmans, Enrico Maria Vitucci, Vittorio Degli-Esposti, Laurent Jacques, Claude Oestges
[10.23919/EuCAP63536.2025.10999736](https://doi.org/10.23919/EuCAP63536.2025.10999736) (2025 19th European Conference on Antennas and Propagation (EuCAP))
- Fully Differentiable Ray Tracing via Discontinuity Smoothing for Radio Network Optimization** Mar 2024
Jérôme Eertmans, Laurent Jacques, Claude Oestges
[10.23919/EuCAP60739.2024.10501570](https://doi.org/10.23919/EuCAP60739.2024.10501570) (2024 18th European Conference on Antennas and Propagation (EuCAP))
- DiffeRT2d - A Differentiable Ray Tracing Python Framework for Radio Propagation** June 2024
Jérôme Eertmans, Claude Oestges, Laurent Jacques
[10.21105/joss.06915](https://doi.org/10.21105/joss.06915) (Journal of Open Source Software)
- Min-Path-Tracing - A Diffraction Aware Alternative to Image Method in Ray Tracing** Mar 2023
Jérôme Eertmans, Claude Oestges, Laurent Jacques
[10.23919/EuCAP57121.2023.10132934](https://doi.org/10.23919/EuCAP57121.2023.10132934) (2023 17th European Conference on Antennas and Propagation (EuCAP))
- Manim Slides - A Python package for presenting Manim content anywhere** Aug 2023
Jérôme Eertmans
[10.21105/jose.00206](https://doi.org/10.21105/jose.00206) (Journal of Open Source Education)

Grants and Awards

Best Propagation Paper Award – Dublin, IE EurAAP – EuCAP 2026.	Apr 2026
TICRA-EurAAP Travel Grant – Dublin, IE EuCAP 2026 travel grant.	Apr 2026
Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (FNRS) – Brussels, BE FRS FNRS <i>Accomplishment of a stay abroad</i> .	Sept 2024 – Dec 2024
Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (WBI) – Brussels, BE WBI <i>Bourses d'Excellence WORLD</i> .	Sept 2024 – Dec 2024
STSM CA20120 - Visit to Pr. Vittorio Degli-Esposti – Brussels, BE Cost Action Short-term Scientific Missions.	Apr 2024
Participation in EuCAP 2024 – Brussels, BE FRS FNRS <i>Participation in a scientific meeting</i> .	Mar 2023

Personal Achievements

ADE Scheduler: An open source alternative to UCLouvain's scheduling tool, that became the official university tool in Sep. 2024.

Manim Slides: An open source presentation tool, used by many for teaching, conference presentations, or thesis defenses.

Scouts: 6 years as a scout (Les Scouts) leader.

Reviews

EuCAP: EuCAP 2026 (5 reviews)

IEEE: Access (5 reviews), Transactions on Vehicular Technology (6 reviews), Antennas and Wireless Propagation Letters (1 review), and Open Journal of the Communications Society (1 review)

Journal of Open Source Software: 1 review (see [#9384](#))

Nature Partner Journals: Wireless Technology: 1 review

Referees

Prof. Claude Oestges, Université catholique de Louvain: claudio.oestges@uclouvain.be

Prof. Laurent Jacques, Université catholique de Louvain: laurent.jacques@uclouvain.be

Prof. Vittorio Degli-Esposti, University of Bologna: v.degliesposti@unibo.it